
GRIFFITH UNIVERSITY

School of Information and Communication Technology

LAB ACTIVITY GUIDE

7015ICT - Cyber Security Operation Centres

Activity 1

DMZ Networks

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This document is prepared for educational purposes within the Griffith University cybersecurity curriculum. All configurations and environments described are intended exclusively for use in isolated, non-production lab settings.

1. Prerequisites

Ensure all prerequisites listed below are in place before starting this activity. Contact your instructor if access or resources are unavailable.

1.1 Required Access

Log in to the Azure Windows host using the following credentials:

```
Username: lab-user
Password: #Griffith2024
```

Before beginning, confirm all the following:

- Access to the Azure Windows host and Hyper-V Manager is working
- All four required VMs are present in Hyper-V Manager: External Gateway, Internal Gateway, Ubuntu Server, Ubuntu Desktop
- VM login credentials are working
- You are working in the lab environment only and not on a production or personal network.

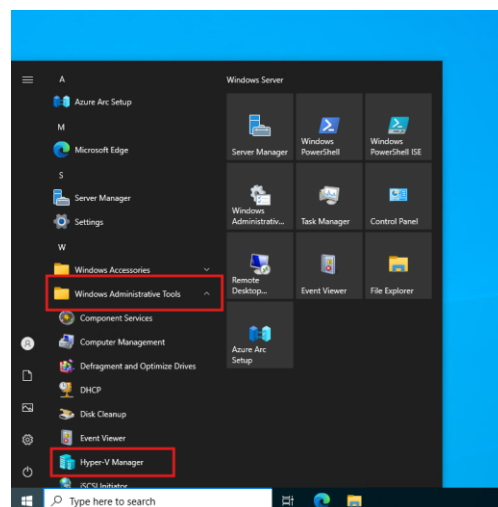


Figure 1. Locating Hyper-V Manager via Windows Administrative Tools in the Start menu

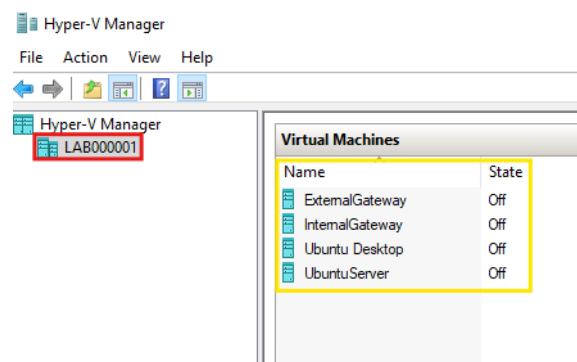


Figure 2. Hyper-V Manager showing all four required VMs on host LAB000001

2. Environment and Topology

2.1 Environment Summary

This activity uses a segmented Hyper-V lab environment hosted on a Microsoft Azure Windows Server VM. The environment consists of an external-facing gateway, a DMZ network hosting a web server, an internal gateway, and an internal client desktop. The DMZ network uses the 192.168.1.0/24 address range, and the internal network uses the 10.10.1.0/24 range. Traffic from the internal and DMZ networks reaches the internet through the External Gateway using NAT.

All VMs are managed through Hyper-V Manager on the Azure Windows host. Three Hyper-V virtual switches are used: ExternalNetwork, DMZNetwork, and InternalNetwork. Each VM is connected to one or more of these switches depending on its role.

2.2 VM and Host Details

VM / Host Name	OS / Role	Network Adapters	Notes
Azure Windows Host	Windows Server / Hyper-V Host	ExternalNetwork, DMZNetwork, InternalNetwork	Hosts and manages the lab VMs in Hyper-V
External Gateway	Ubuntu Server / External Gateway Router	ExternalNetwork, DMZNetwork	Provides outbound internet access and NAT
Ubuntu Server	Ubuntu Server / DMZ Web Server	DMZNetwork	Hosts the web server in the DMZ segment
Internal Gateway	Ubuntu Server / Internal Gateway Router	DMZNetwork, InternalNetwork	Routes traffic between the internal and DMZ networks
Ubuntu Desktop	Ubuntu Desktop / Internal Host	InternalNetwork	Client workstation used to test connectivity

2.3 Network Topology Diagram

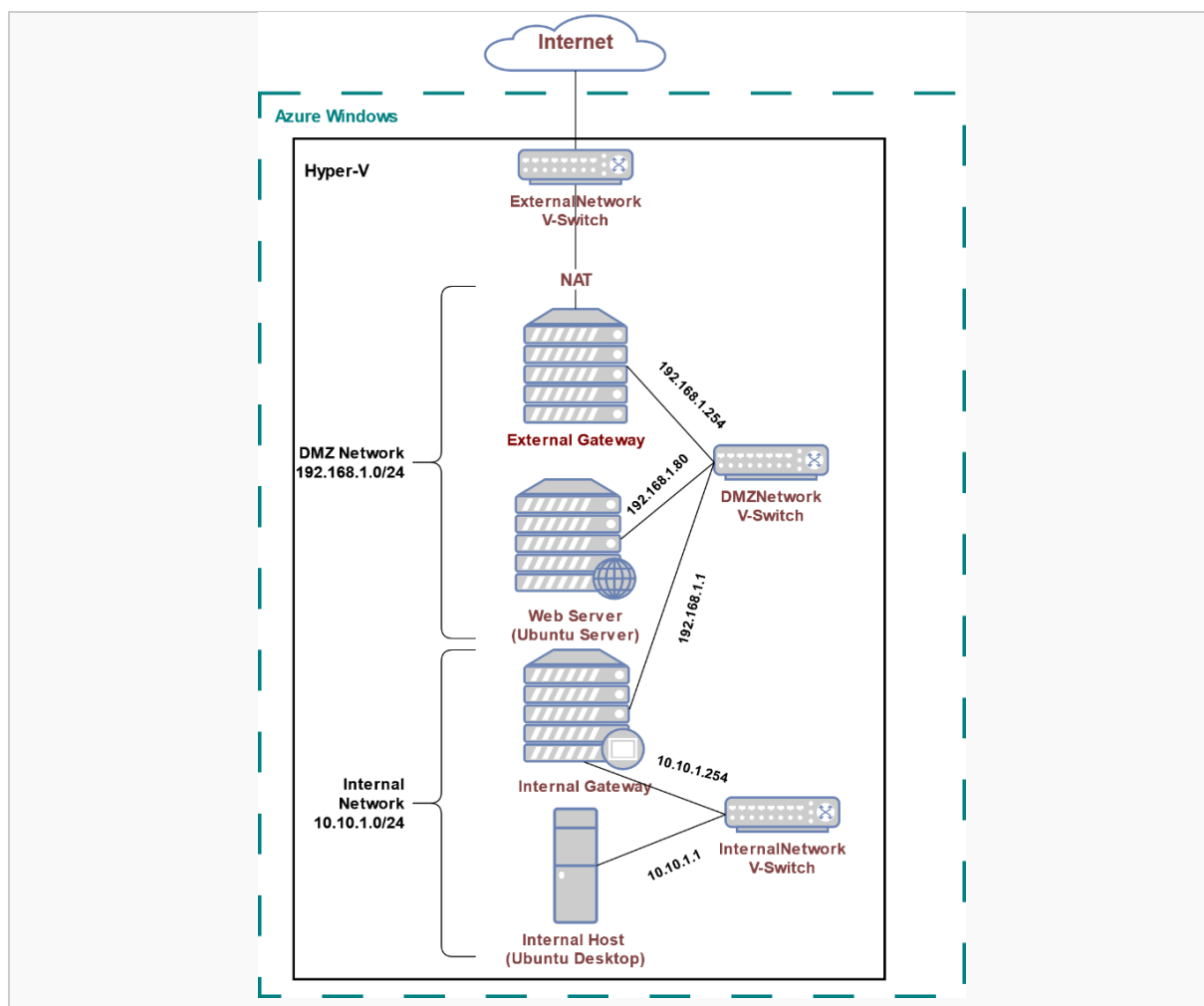


Figure 3. Network topology diagram showing all segments, interfaces, and IP addressing

2.4 IP Addressing Plan

Host / Interface	IP Address	Subnet Mask	Default Gateway	Segment
External Gateway - eth0	DHCP-assigned	DHCP-assigned	DHCP-assigned	External Network
External Gateway - eth1	192.168.1.254	255.255.255.0	N/A	DMZ
Internal Gateway - eth0	192.168.1.1	255.255.255.0	192.168.1.254	DMZ
Internal Gateway - eth1	10.10.1.254	255.255.255.0	N/A	Internal
Ubuntu Server - eth0	192.168.1.80	255.255.255.0	192.168.1.254	DMZ
Ubuntu Desktop - eth0	10.10.1.1	255.255.255.0	10.10.1.254	Internal

3. Procedure

Follow the steps below in the order presented. Read each step in full before executing. Where verification checkpoints appear, confirm the expected result before proceeding to the next phase.

Warning: Perform all configurations only within the designated lab environment. Do not connect lab VMs to the internet or production networks outside of the External Gateway's controlled NAT path.

3.1 Verify VM Availability in Hyper-V

Before configuring any network settings, confirm that all required VMs are present in Hyper-V Manager and that you can log in to each one successfully.

Step-by-step Instructions:

1. Log in to the Azure Windows host using the credentials in Section 1.1.
2. Open the Start menu, expand Windows Administrative Tools, and select Hyper-V Manager.
3. In the Virtual Machines pane, confirm all four VMs are listed:
 - External Gateway
 - Internal Gateway
 - Ubuntu Server
 - Ubuntu Desktop.
4. Start each VM in turn:
 - Right-click a VM and select Start.
 - Once running, right-click again and select Connect.
5. Log in to each VM using the following credentials:

```
Username: user
Password: password
```

A screenshot of a terminal window showing the login process for an Ubuntu Server. The prompt is 'InternalGateway login: user'. The user enters 'user' for the username and 'password' for the password. The terminal output shows the Ubuntu 24.04 LTS welcome message, system information, and a list of available updates. A yellow box highlights the login credentials: 'Username: user' and 'Password: password'.

```
InternalGateway login: user
Password:
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.8.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Mar 27 09:54:27 AM UTC 2026

System load: 0.08      Memory usage: 10%    Processes:   98
Usage of /: 34.6% of 12.25GB  Swap usage: 0%      Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

user@internalgateway:~$
```

Figure 4. Ubuntu Server CLI login prompt confirming successful authentication with provided credentials

On each VM, run the following command to confirm the currently detected network interfaces and their names before editing any configuration:

```
ip a
```

(For Ubuntu Desktop VM, open “Terminal” to run the command)

Note the interface names shown (typically eth0 for the first adapter and eth1 for the second). Use these names when editing the Netplan configuration files in Section 3.2.

3.2 Configure Virtual Network Adapters in Hyper-V

- The Internal Gateway must be connected to both DMZNetwork and InternalNetwork adapters.
- The External Gateway must be connected to ExternalNetwork and DMZNetwork adapters.
- Ubuntu Server and Ubuntu Desktop each use a single adapter.

Warning: Verify and configure all virtual switch assignments before editing any IP configuration files. An incorrect switch assignment will cause connectivity failures that are difficult to diagnose after IP settings have been applied.

Internal Gateway - Assign Network Adapters:

1. In Hyper-V Manager, right-click Internal Gateway and select Settings.
2. In the Hardware list on the left, select Add Hardware.
3. In the right pane, select Network Adapter from the list and click Add.
4. Select the newly added Network Adapter entry in the left-hand list.
5. In the Virtual switch dropdown, select InternalNetwork. Click Apply, then OK.
6. Repeat the steps 2-5 to add DMZNetwork adapter.
7. Confirm that Internal Gateway now has two Network Adapter entries: one assigned to DMZNetwork and one assigned to InternalNetwork.

(By default, the network adapter is assigned to ExternalNetwork.)

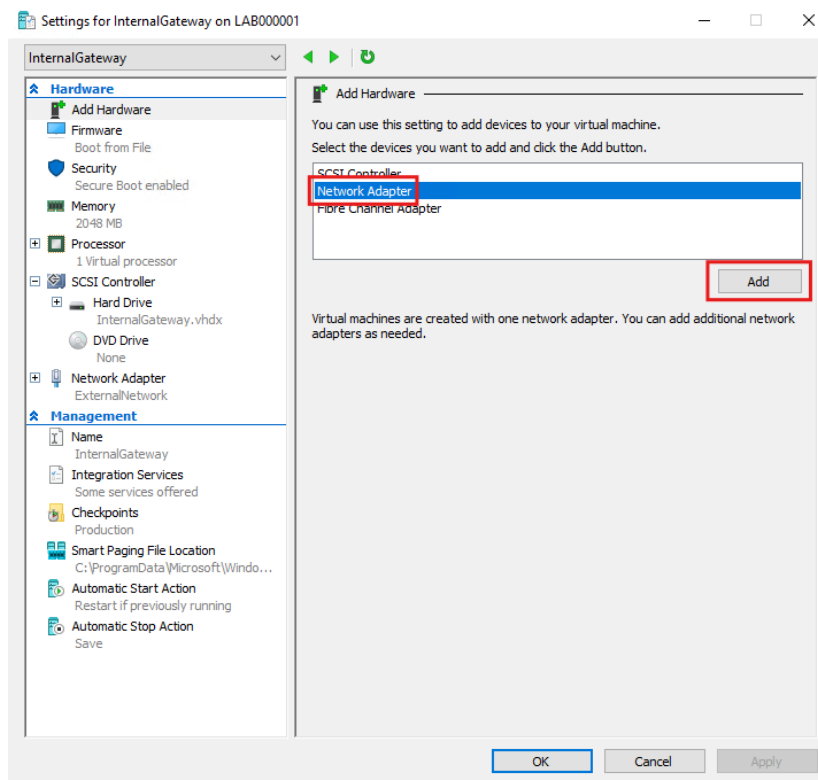


Figure 5. Internal Gateway settings - selecting Network Adapter under Add Hardware before clicking Add

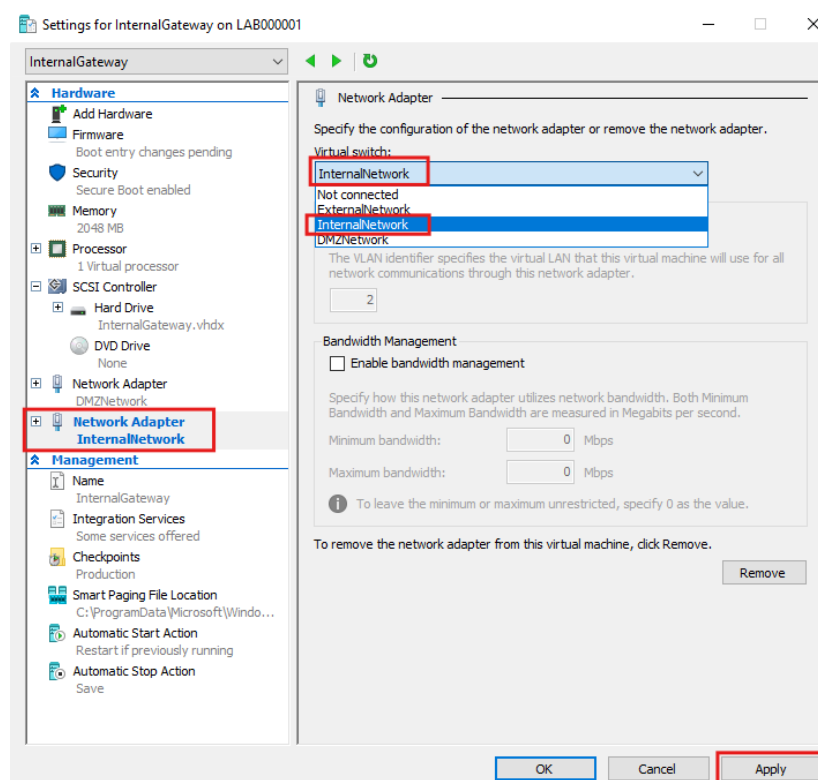


Figure 6. Internal Gateway settings - second adapter assigned to InternalNetwork virtual switch

External Gateway - Add Network Adapters:

1. In Hyper-V Manager, right-click External Gateway and select Settings.
2. Follow same steps from Internal Gateway to add Network Adapter, confirm Network Adapter entries are present in the left-hand list.
3. Assign the adapters to ExternalNetwork and DMZNetwork.
4. Click Apply, then OK.

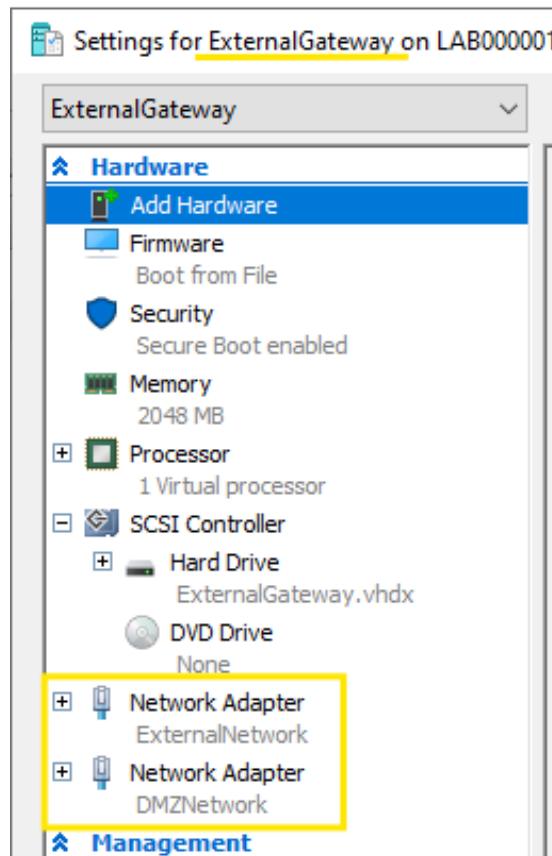


Figure 7. External Gateway settings showing two network adapters assigned to ExternalNetwork and DMZNetwork

Ubuntu Desktop and Ubuntu Server - Verify Adapter Assignment:

- Ubuntu Desktop must be connected to Internal Network only.
- Ubuntu Server must be connected to DMZNetwork only.
- Check each VM's settings in Hyper-V Manager and correct any incorrect switch assignments before continuing.

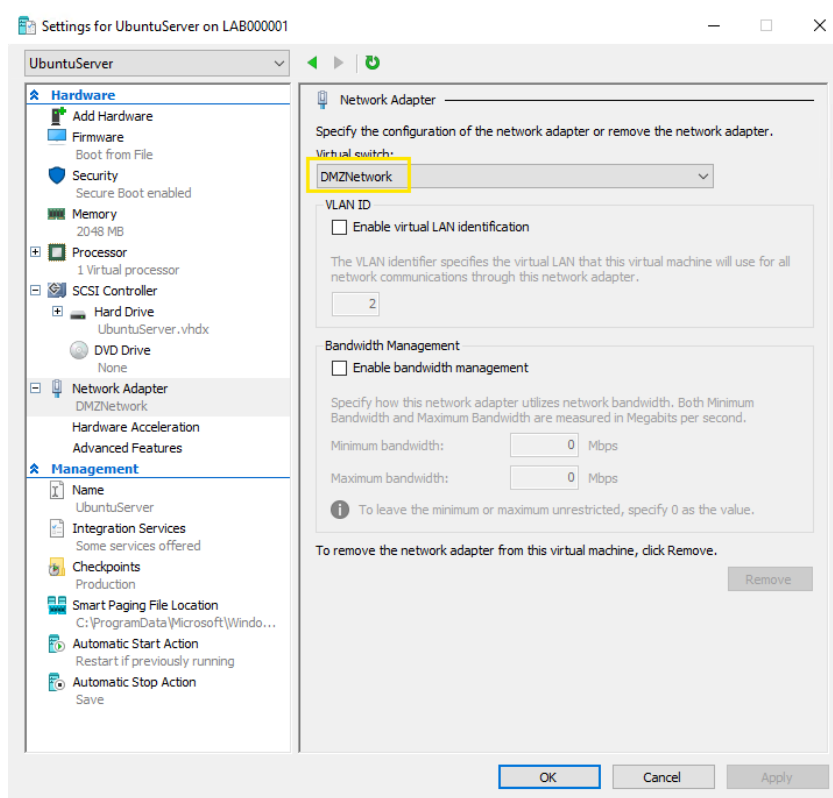


Figure 8. Ubuntu Server Hyper-V settings - Network Adapter settings

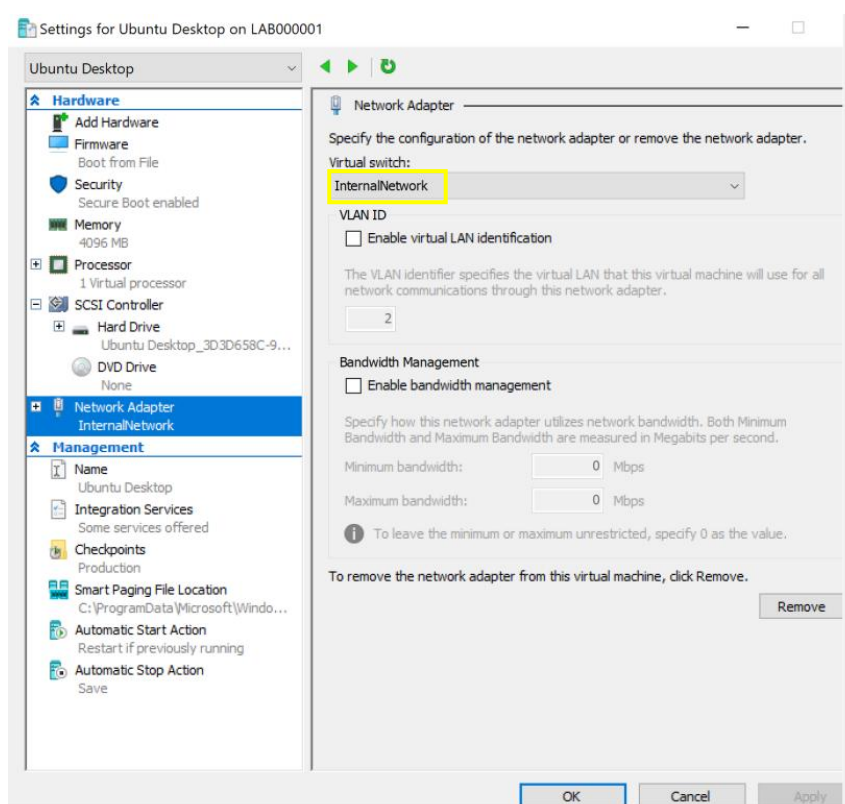


Figure 9. Ubuntu Desktop Hyper-V settings - Network Adapter settings

3.3 Configure Static IP Addressing - Server VMs

The three server VMs (Internal Gateway, External Gateway, and Ubuntu Server) use Netplan for network configuration. Edit the Netplan configuration file on each VM using the values from the IP addressing plan in Section 2.4.

Warning: Netplan configuration files use YAML syntax. Incorrect indentation will cause the configuration to fail. Use exactly two spaces for each level of indentation. Do not use tab.

Internal Gateway

1. Log in to the Internal Gateway VM.
2. Confirm the Netplan filename:

```
ls /etc/netplan
```

3. Open the configuration file for editing:

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

(The password to all the required access within the VMs is 'password'.)

4. Replace the file contents with the following configuration:

```
network:
  version: 2
  ethernets:
    eth0:
      dhcp4: no
      addresses:
        - 192.168.1.1/24
      routes:
        - to: 0.0.0.0/0
          via: 192.168.1.254
      nameservers:
        addresses:
          - 8.8.8.8
    eth1:
      dhcp4: no
      addresses:
        - 10.10.1.254/24
```

5. Save the file (Ctrl+O, then Enter) and exit nano (Ctrl+X).
6. Apply the configuration:

```
sudo netplan apply
```

7. Verify both interfaces have the correct addresses:

```
ip addr show eth0
ip addr show eth1
```

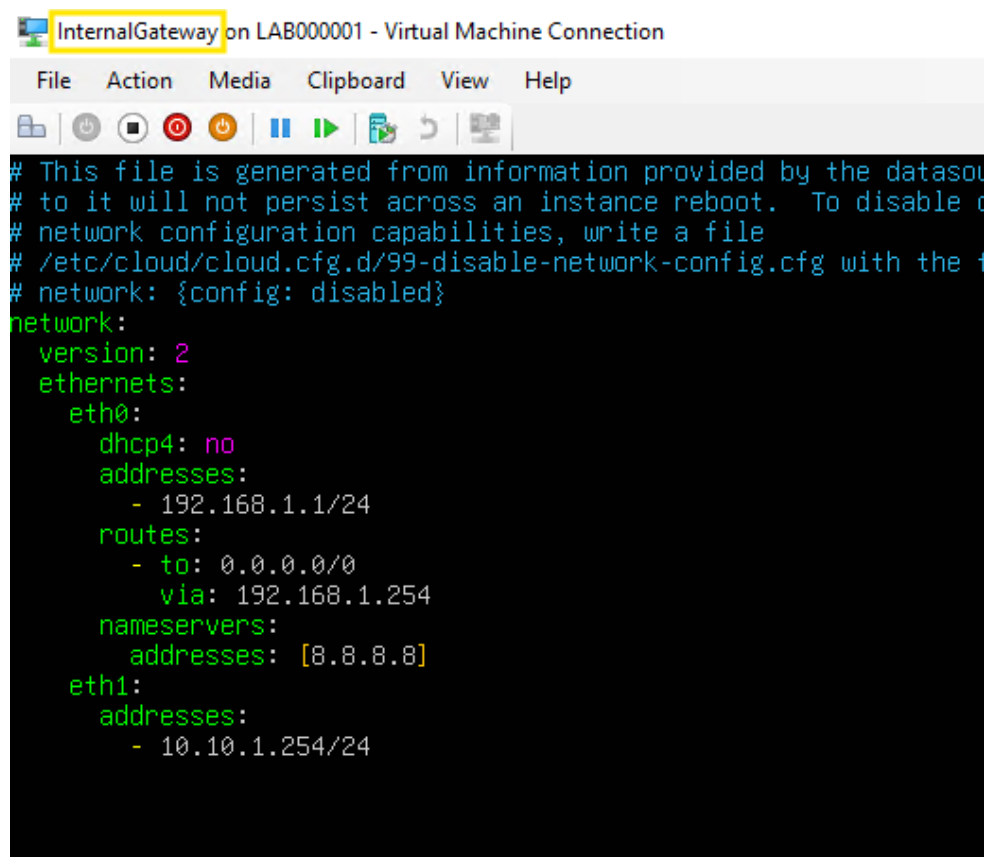


Figure 10. Internal Gateway Netplan configuration - both eth0 (DMZ) and eth1 (Internal) configured with static addresses

****Important****

After configuring and finishing with each VM, save the VM for future activities. This prevents you from having to configure the VMs from the beginning.

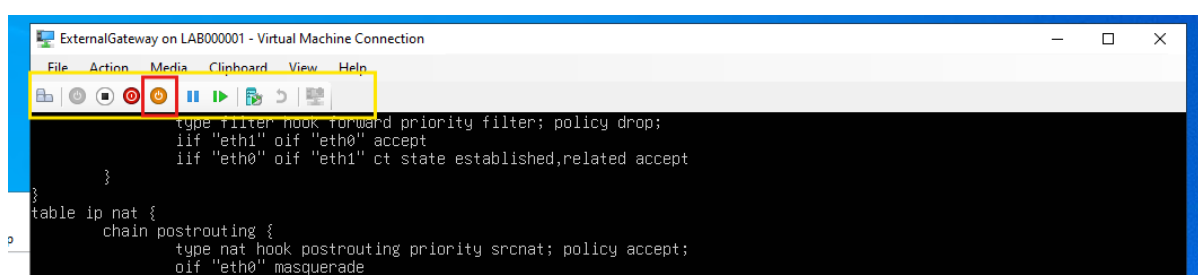


Figure 11. Press the Save button (red box)

External Gateway

1. Log in to the External Gateway VM.
2. Open the configuration file for editing:

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

3. Replace the file contents with the following configuration:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    eth0:
      dhcp4: true
    eth1:
      dhcp4: no
      addresses: [192.168.1.254/24]
      routes:
        - to: 10.10.1.0/24
          via: 192.168.1.1
```

4. Save and exit (Ctrl+O, Enter, Ctrl+X), then apply:

```
sudo netplan apply
```

5. Verify the interfaces:

```
ip addr show eth0
ip addr show eth1
```

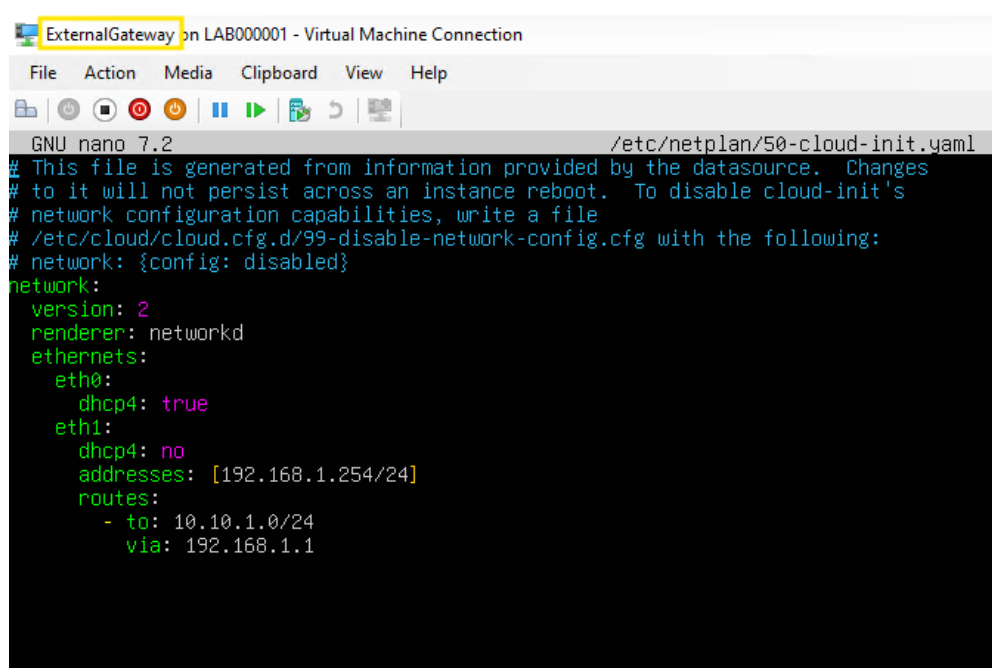


Figure 11. External Gateway Netplan configuration - eth0 on DHCP, eth1 with static DMZ address and internal route

Ubuntu Server

1. Log in to the Ubuntu Server VM.
2. Open the configuration file for editing:

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

3. Replace the file contents with the following configuration:

```
network:
  version: 2
  ethernet:
    eth0:
      dhcp4: no
      addresses:
        - 192.168.1.80/24
      routes:
        - to: 0.0.0.0/0
          via: 192.168.1.254
      nameservers:
        addresses: [8.8.8.8]
```

4. Save and exit, then apply:

```
sudo netplan apply
```

5. Verify the interface:

```
ip addr show eth0
```

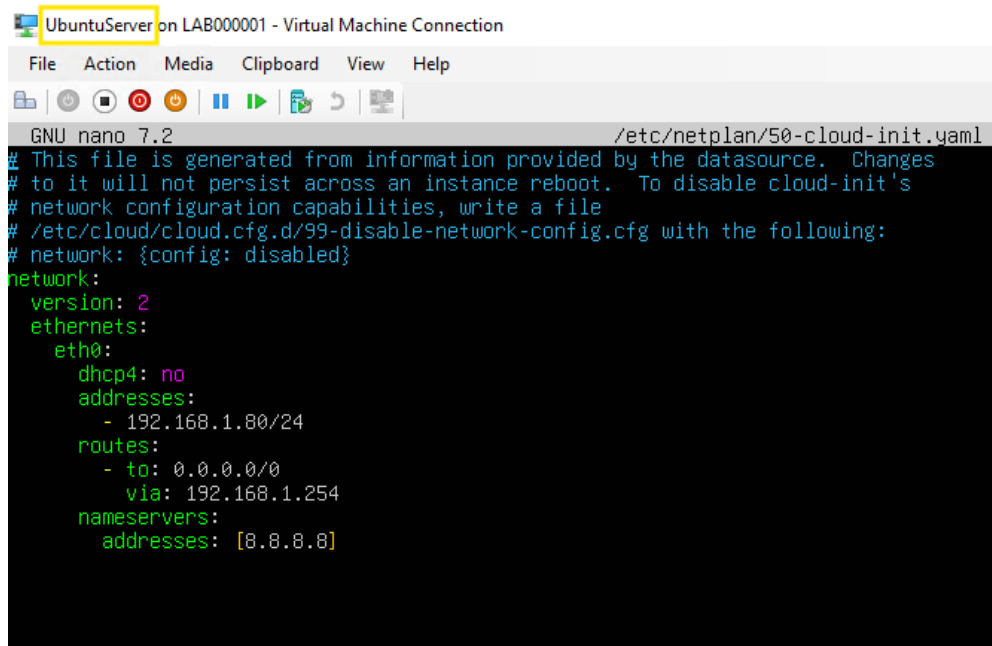


Figure 12. Ubuntu Server Netplan configuration - static address 192.168.1.80/24 with default route via External Gateway

3.4 Configure Ubuntu Desktop Network Settings

Ubuntu Desktop uses the GNOME network settings GUI rather than Netplan for IP configuration. The steps below configure a static IPv4 address on the wired connection.

Step-by-step Instructions:

1. On the Ubuntu Desktop VM, click the network icon in the top-right corner of the taskbar to open the quick settings panel.

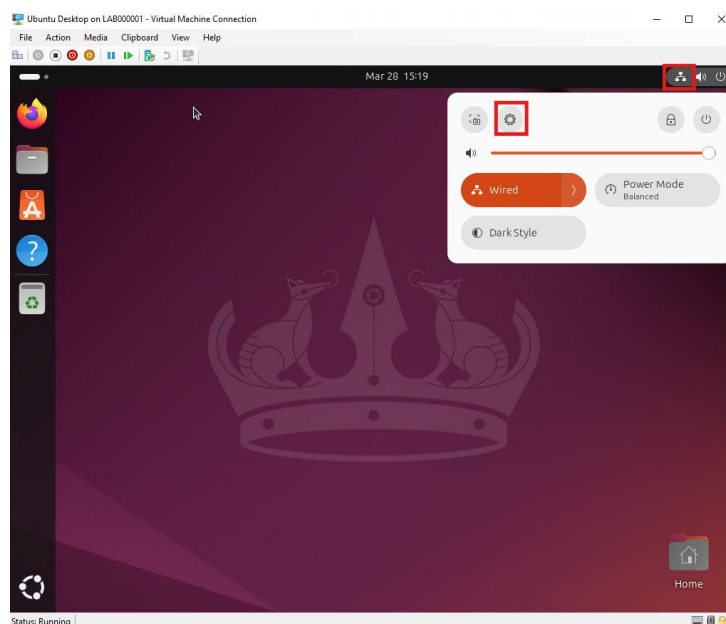


Figure 13. Ubuntu Desktop quick settings panel - click the settings gear icon (highlighted) to open full network settings

2. In the quick settings panel, click the settings gear icon next to Wired to open the network settings window.

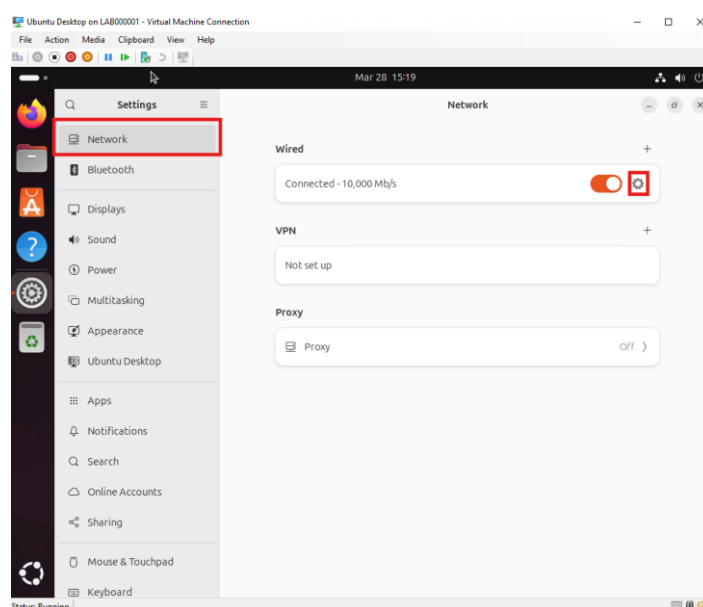


Figure 14. Ubuntu Desktop network settings showing the Wired connection is active - click the gear icon to edit

3. Select the IPv4 tab.
4. Change the IPv4 Method to Manual.
5. Enter the following values in the Addresses section:

```
Address: 10.10.1.1  
Netmask: 255.255.255.0  
Gateway: 10.10.1.254
```

6. In the DNS field enter:

```
8.8.8.8
```

7. Click Apply. Close and reopen the network settings window to confirm the address has been saved.

(If the connection does not update immediately, toggle the wired connection off and back on using the toggle switch in the network settings window.)

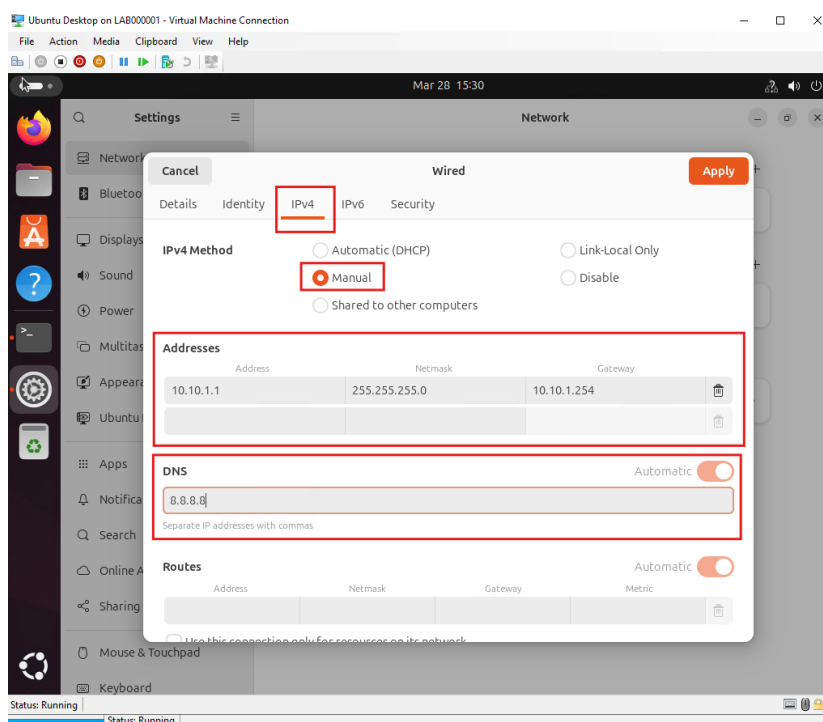


Figure 15. Ubuntu Desktop IPv4 Manual configuration

3.4.1 Verification Checkpoint 3.4

Open a terminal on Ubuntu Desktop and run the following commands to confirm the configuration:

```
ip a  
ip route
```

3.5 Enable IP Forwarding on Both Gateway VMs

IP forwarding must be enabled on both the Internal Gateway and External Gateway so that each VM can pass traffic between its two network interfaces. Without this setting, a multi-homed Linux system will not route traffic between connected subnets.

Perform the following steps on both the Internal Gateway and the External Gateway.

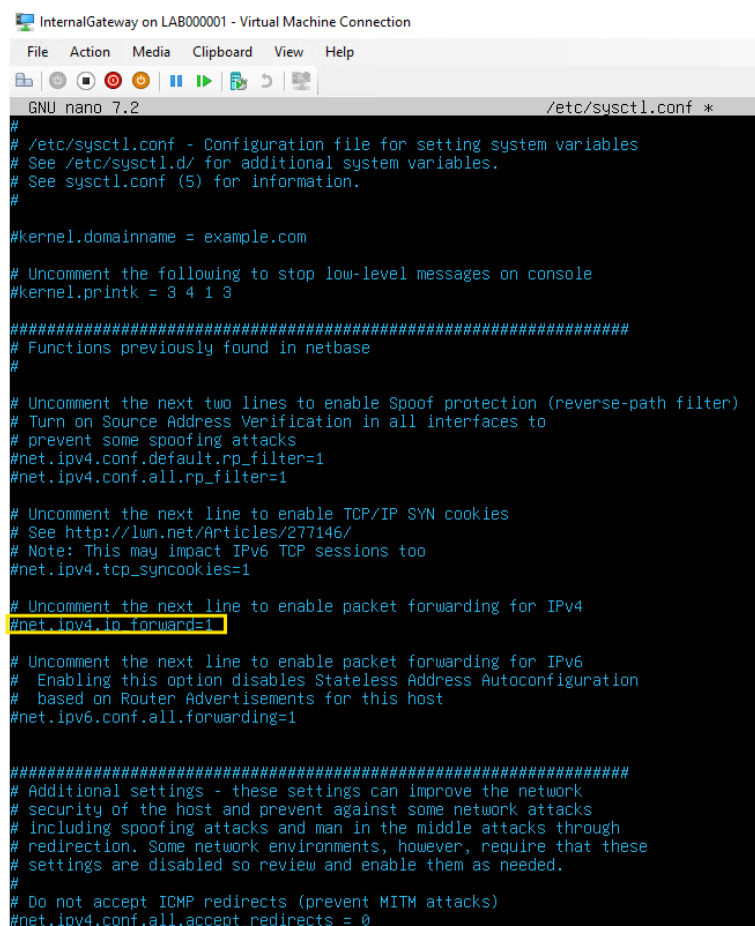
Step-by-step Instructions:

1. Log in to the gateway VM.
2. Open the kernel networking settings file:

```
sudo nano /etc/sysctl.conf
```

3. Locate the following line (it will be commented out with a # symbol):

```
#net.ipv4.ip_forward=1
```

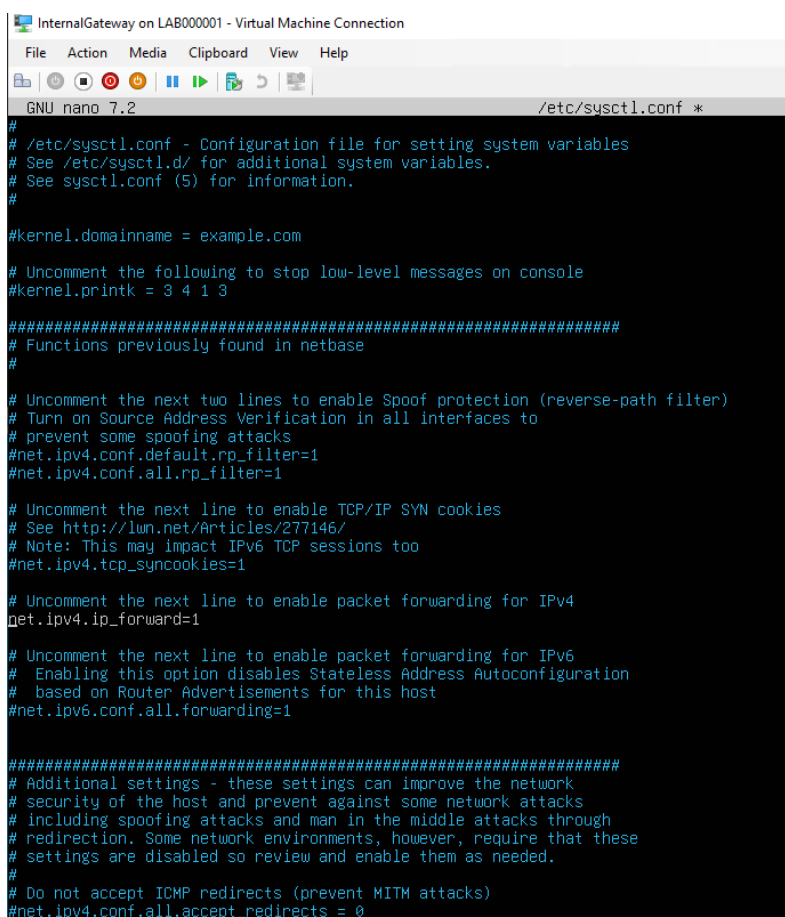


```
InternalGateway on LAB000001 - Virtual Machine Connection
File Action Media Clipboard View Help
GNU nano 7.2 /etc/sysctl.conf *
#
# /etc/sysctl.conf - Configuration file for setting system variables
# See /etc/sysctl.d/ for additional system variables.
# See sysctl.conf (5) for information.
#
#kernel.domainname = example.com
#
# Uncomment the following to stop low-level messages on console
#kernel.printk = 3 4 1 3
#####
# Functions previously found in netbase
#
# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1
#
# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1
#
# Uncomment the next line to enable packet forwarding for IPv4
#net.ipv4.ip_forward=1
#
# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
#####
# Additional settings - these settings can improve the network
# security of the host and prevent against some network attacks
# including spoofing attacks and man in the middle attacks through
# redirection. Some network environments, however, require that these
# settings are disabled so review and enable them as needed.
#
# Do not accept ICMP redirects (prevent MITM attacks)
#net.ipv4.conf.all.accept_redirects = 0
```

Figure 16. sysctl.conf before editing

4. Remove the # character to uncomment the line, so it reads exactly:

```
net.ipv4.ip_forward=1
```

```

InternalGateway on LAB000001 - Virtual Machine Connection
File Action Media Clipboard View Help
GNU nano 7.2 /etc/sysctl.conf *
#
# /etc/sysctl.conf - Configuration file for setting system variables
# See /etc/sysctl.d/ for additional system variables.
# See sysctl.conf (5) for information.
#
#kernel.domainname = example.com
# Uncomment the following to stop low-level messages on console
#kernel.printk = 3 4 1 3
#####
# Functions previously found in netbase
#
# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1
#
# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1
#
# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
#
# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
#####
# Additional settings - these settings can improve the network
# security of the host and prevent against some network attacks
# including spoofing attacks and man in the middle attacks through
# redirection. Some network environments, however, require that these
# settings are disabled so review and enable them as needed.
#
# Do not accept ICMP redirects (prevent MITM attacks)
#net.ipv4.conf.all.accept_redirects = 0

```

Figure 17. sysctl.conf after editing

5. Save and exit (Ctrl+O, Enter, Ctrl+X), then reload the configuration:

```
sudo sysctl -p
```

6. Verify the setting is active:

```
sysctl net.ipv4.ip_forward
```

Expected output:

```
net.ipv4.ip_forward = 1
```

7. Repeat Steps 1 through 6 on External gateway VM.

3.6 Configure NAT and Forwarding on External Gateway using nftables

This activity uses nftables to implement NAT and packet forwarding on the External Gateway. nftables is the modern successor to iptables and is the preferred firewall and NAT framework on current Linux systems. The ruleset configured below replaces the older iptables-based approach while achieving the same network behaviour.

3.6.1 Install nftables

On the External Gateway VM, run the following commands to update the package list and install nftables:

```
sudo apt update
sudo apt install nftables -y
```

3.6.2 Enable the nftables Service

```
sudo systemctl enable nftables
sudo systemctl start nftables
```

3.6.3 Create the nftables Ruleset

1. Open the nftables configuration file:

```
sudo nano /etc/nftables.conf
```

2. Replace the entire file contents with the following ruleset:

```
#!/usr/sbin/nft -f

flush ruleset

table inet filter {
    chain forward {
        type filter hook forward priority 0;
        policy drop;

        iif "eth1" oif "eth0" accept
        iif "eth0" oif "eth1" ct state established,related accept
    }
}

table ip nat {
    chain postrouting {
        type nat hook postrouting priority 100;
        oif "eth0" masquerade
    }
}
```

3. Alternatively, you can download the configuration file straight from Github:

```
wget "https://raw.githubusercontent.com/rishonmathew/griffith-assessment-
automarker/main/activity1/external-gateway-nftables.conf"

sudo cp external-gateway-nftables.conf /etc/nftables.conf

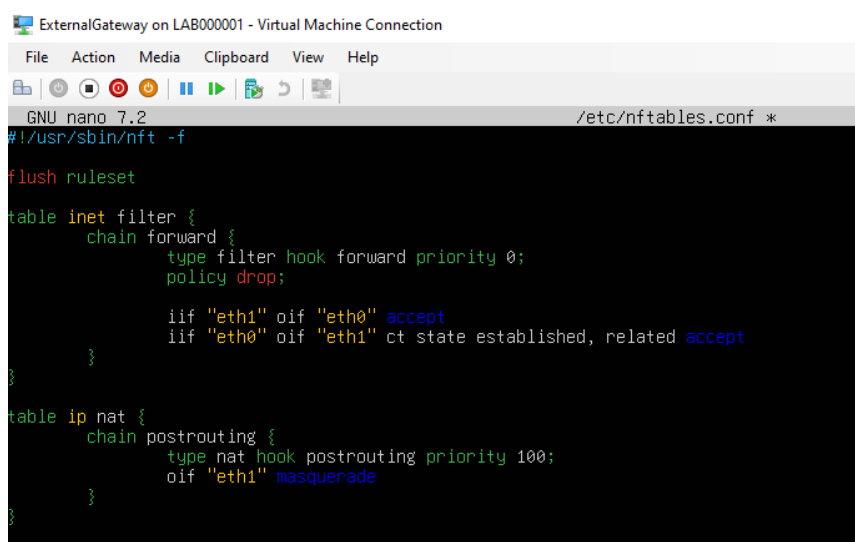
sudo nft -f /etc/nftables.conf

sudo systemctl restart nftables
```

This ruleset performs the following functions:

- Allows forwarded traffic from the DMZ side (eth1) to the external side (eth0)
- Allows return traffic from the external side back to the DMZ when it is part of an established or related connection
- Applies source NAT masquerading on all outbound traffic leaving eth0

4. Save and exit the file (Ctrl+O, Enter, Ctrl+X):



```

GNU nano 7.2 /etc/nftables.conf *
#!/usr/sbin/nft -f

flush ruleset

table inet filter {
    chain forward {
        type filter hook forward priority 0;
        policy drop;

        iif "eth1" oif "eth0" accept
        iif "eth0" oif "eth1" ct state established, related accept
    }
}

table ip nat {
    chain postrouting {
        type nat hook postrouting priority 100;
        oif "eth1" masquerade
    }
}

```

Figure 18. nftables configuration file on External Gateway

3.6.4 Apply the Ruleset

```
sudo nft -f /etc/nftables.conf
```

3.6.5 Verify the Ruleset

```
sudo nft list ruleset
```

Expected output should include:

- A table inet filter section containing the forward chain with two accept rules
- A table ip nat section containing the postrouting chain with a masquerade rule applied to eth0

```

user@externalgateway:~$ sudo nft list ruleset
[sudo] password for user:
table inet filter {
    chain forward {
        type filter hook forward priority filter; policy drop;
        iif "eth1" oif "eth0" accept
        iif "eth0" oif "eth1" ct state established,related accept
    }
}
table ip nat {
    chain postrouting {
        type nat hook postrouting priority srcnat; policy accept;
        oif "eth0" masquerade
    }
}

```

Figure 19. Verified Ruleset

3.6.6 Persistence

Because the rules are stored in `/etc/nftables.conf` and the `nftables` service is enabled, the ruleset will be automatically restored on every system reboot. No additional steps are needed to make these rules persistent.

4. Expected Results

Upon successful completion of this activity, the following outcomes should be observable across all VMs:

VM	ip a should show	ip route should show
External Gateway	eth0 with DHCP external IP, eth1 = 192.168.1.254/24	default via external network, route to 10.10.1.0/24 via 192.168.1.1
Internal Gateway	eth0 = 192.168.1.1/24, eth1 = 10.10.1.254/24	default via 192.168.1.254, connected routes for both subnets
Ubuntu Server	eth0 = 192.168.1.80/24	default via 192.168.1.254
Ubuntu Desktop	eth0 = 10.10.1.1/24	default via 10.10.1.254

4.1 Verified Interface and Routing Output

The screenshots below show confirmed `ip a` and `ip route` output from each VM after successful configuration.

```

user@internalgateway:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:00:07:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.1/24 brd 192.168.1.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe00:705/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:00:07:07 brd ff:ff:ff:ff:ff:ff
    inet 10.10.1.254/24 brd 10.10.1.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe00:707/64 scope link
        valid_lft forever preferred_lft forever
user@internalgateway:~$ ip route
default via 192.168.1.254 dev eth0 proto static
10.10.1.0/24 dev eth1 proto kernel scope link src 10.10.1.254
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.1
  
```

Figure 20. Internal Gateway `ip a` and `ip route` output - `eth0` = 192.168.1.1/24 (DMZ), `eth1` = 10.10.1.254/24 (Internal), default route via 192.168.1.254

```

user@externalgateway:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:00:07:04 brd ff:ff:ff:ff:ff:ff
    inet 172.16.10.101/24 metric 100 brd 172.16.10.255 scope global dynamic eth0
        valid_lft 690701sec preferred_lft 690701sec
    inet6 fe80::215:5dff:fe00:704/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:00:07:08 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.254/24 brd 192.168.1.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe00:708/64 scope link
        valid_lft forever preferred_lft forever
user@externalgateway:~$ ip route
default via 172.16.10.254 dev eth0 proto dhcp src 172.16.10.101 metric 100
10.10.1.0/24 via 192.168.1.1 dev eth1 proto static
168.63.129.16 via 172.16.10.254 dev eth0 proto dhcp src 172.16.10.101 metric 100
172.16.10.0/24 dev eth0 proto kernel scope link src 172.16.10.101 metric 100
172.16.10.254 dev eth0 proto dhcp scope link src 172.16.10.101 metric 100
192.168.1.0/24 dev eth1 proto kernel scope link src 192.168.1.254

```

Figure 21. External Gateway ip a and ip route output - eth0 DHCP external IP, eth1 = 192.168.1.254/24 (DMZ), route to 10.10.1.0/24 via 192.168.1.1

```

user@server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:00:07:06 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.80/24 brd 192.168.1.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe00:706/64 scope link
        valid_lft forever preferred_lft forever
user@server:~$ ip route
default via 192.168.1.254 dev eth0 proto static
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.80

```

Figure 22. Ubuntu Server ip a and ip route output - eth0 = 192.168.1.80/24 (DMZ), default route via 192.168.1.254

In addition to the interface and routing checks above, all the following should be true at the end of the activity:

- All four VMs are correctly connected to their intended Hyper-V virtual switches
- Internal Gateway successfully routes traffic between the internal (10.10.1.0/24) and DMZ (192.168.1.0/24) networks
- External Gateway provides NAT for outbound internet access from internal and DMZ hosts
- Ubuntu Desktop can ping the Internal Gateway (10.10.1.254) and reach DMZ hosts (e.g. 192.168.1.80)
- Ubuntu Desktop can access the internet through a browser, curl, or wget
- External Gateway can access the internet directly via its DHCP-assigned external address

Note: If your results differ from those described above, refer to Section 5: Troubleshooting before contacting your instructor. Document any unexpected behaviour with screenshots and command output.

5. Troubleshooting

The table below lists common issues encountered during this activity, along with their likely causes and recommended resolution steps.

Symptom / Issue	Likely Cause	Resolution
Ubuntu Desktop cannot ping 10.10.1.254	Ubuntu Desktop connected to wrong Hyper-V virtual switch	Open Hyper-V settings and confirm Ubuntu Desktop is connected to InternalNetwork
Internal Gateway shows only one NIC after setup	Second network adapter not added correctly in Hyper-V	Re-open Hyper-V settings for Internal Gateway and add or verify the second network adapter
sudo netplan apply returns a YAML error	Incorrect indentation or a mistyped IP address in the configuration file	Re-open the netplan file and check indentation carefully. All nested keys must use two-space indentation. Verify all IP addresses match the values in the IP addressing plan
Correct IP is shown but cannot reach the other subnet	IP forwarding not enabled on one or both gateway VMs	Run <code>sysctl net.ipv4.ip_forward</code> on Internal Gateway and External Gateway and confirm the value is 1 on each
Internal hosts can reach the gateway but not the internet	nftables rules not applied or incorrectly configured	Run <code>sudo nft list ruleset</code> and verify that the forward chain and postrouting chain are present. If missing, re-apply with <code>sudo nft -f /etc/nftables.conf</code>
sudo nft -f /etc/nftables.conf returns an error	Syntax issue in the nftables configuration file	Re-open /etc/nftables.conf and check spacing, braces, and interface names. Ensure iif and oif values match the actual interface names shown by <code>ip a</code>
Internet works until reboot	nftables service not enabled	Run <code>sudo systemctl enable nftables</code> and confirm the service is active with <code>sudo systemctl status nftables</code>
Ubuntu Desktop cannot reach external destinations	IP forwarding not enabled on one or both gateway VMs	Run <code>sysctl net.ipv4.ip_forward</code> on Internal Gateway and External Gateway and confirm the value is 1
Firefox works but ping 8.8.8.8 fails	ICMP blocked upstream by the cloud provider or lab firewall	Use Firefox, curl, or wget to verify internet connectivity instead of relying on ping alone

Tip: Always check interface status with `ip a` and routing with `ip route` before editing configuration files. Most connectivity failures are caused by an incorrect virtual switch assignment in Hyper-V or a YAML indentation error in Netplan - not by a misconfigured IP address.

6. Automarker Guide

6.1 Overview

The Activity 1 automarker is a Bash script that runs on any of the four lab VMs and automatically validates the network configuration for the DMZ Networks activity. It detects which VM it is on, runs the appropriate checks, and reports colour-coded pass/fail results with numbered error codes.

This guide covers everything needed to: install prerequisites, download the script onto each VM, run it, and interpret the results.

⚠ Warning: Complete the full Activity 1 procedure before running the automarker. The External Gateway must have its nftables masquerade rule and IP forwarding configured before other VMs can reach the internet.

6.2 Prerequisites — Install curl on All VMs

The automarker uses `curl` to test internet connectivity. Run the following on **every VM** before downloading or running the script:

```
sudo apt update
sudo apt install curl -y
```

Repeat this on: External Gateway, Internal Gateway, Ubuntu Server, and Ubuntu Desktop.

The script must be run with `sudo`. Default credentials for all VMs:

```
Username: user
Password: password
```

6.3 Downloading and Running the Script

Each VM requires a different download method depending on whether it has direct internet access. Follow the steps for each VM in the order below.

VM	Download method	Reason
External Gateway	wget from GitHub	Always has direct internet via DHCP on eth0
Internal Gateway	wget from External Gateway (python3 server)	No direct internet — on DMZ only
Ubuntu Server	wget from External Gateway (python3 server)	No direct internet — on DMZ only
Ubuntu Desktop	wget from GitHub	Has internet via both gateways once activity is complete

6.3.1 External Gateway

The External Gateway always has internet access. Download and run directly from GitHub:

```
wget "https://raw.githubusercontent.com/rishonmathew/griffith-  
assessment-automarker/main/activity1/automark_activity1.sh"  
chmod +x automark_activity1.sh  
sudo ./automark_activity1.sh
```

Leave this terminal open after the script finishes — you will need it in Step 3.2 to serve the file to other VMs.

6.3.2 Internal Gateway and Ubuntu Server

These VMs are on the DMZ network only and cannot reach GitHub directly. Instead, serve the script from External Gateway over the local DMZ network using Python's built-in HTTP server.

Step 1 — On External Gateway, start the file server:

```
cd /home/user  
python3 -m http.server 8080
```

Leave this running. You will see log lines appear as the other VMs download the file.

Step 2 — On Internal Gateway, download from External Gateway:

```
wget http://192.168.1.254:8080/automark_activity1.sh  
chmod +x automark_activity1.sh  
sudo ./automark_activity1.sh
```

Step 3 — On Ubuntu Server, download from External Gateway:

```
wget http://192.168.1.254:8080/automark_activity1.sh  
chmod +x automark_activity1.sh  
sudo ./automark_activity1.sh
```

Once both VMs have downloaded the file you can stop the python3 server on External Gateway with `Ctrl+C`.

6.3.3 Ubuntu Desktop

Ubuntu Desktop has internet access through both gateways once the activity is complete.

Download directly from GitHub:

```
wget "https://raw.githubusercontent.com/rishonmathew/griffith-  
assessment-automarker/main/activity1/automark_activity1.sh"  
chmod +x automark_activity1.sh  
sudo ./automark_activity1.sh
```


6.4 Understanding the Output

Indicator	Meaning
[PASS]	Check succeeded — configuration is correct for this item
[FAIL]	Check failed — see the error code printed below it for resolution steps
[INFO]	Informational message — no action required

When a check fails, the script automatically prints relevant live state (current IP addresses, routes, or nftables rules) immediately below the failure to help diagnose the issue without running extra commands.

```

Activity 1 Automarker — DMZ Networks
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VM detected: External Gateway

--- nftables Forward Rules ---
[PASS] nftables forward rule: iif eth1 oif eth0 accept
[PASS] nftables forward rule: iif eth0 oif eth1 ct state
established,related accept

--- Summary ---
Passed : 14          Failed : 0
All checks passed! Activity 1 configuration looks correct.
```

6.5 Troubleshooting

6.5.1 Error E1 — IP address not configured correctly

Re-open the Netplan configuration and verify the address matches the IP plan in the activity guide:

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

Use exactly 2 spaces per indentation level — no tabs. After saving:

```
sudo netplan apply
```

6.5.2 Error E2 — Network interface missing

Open Hyper-V Manager → right-click the VM → Settings → Add Hardware → Network Adapter. Required assignments:

- **External Gateway:** ExternalNetwork and DMZNetwork
- **Internal Gateway:** DMZNetwork and InternalNetwork
- **Ubuntu Server:** DMZNetwork only
- **Ubuntu Desktop:** InternalNetwork only

⚠ Note: After fixing a virtual switch assignment in Hyper-V, no reboot is needed. Re-run the automarker immediately.

6.5.3 Error E3 — Routing or IP forwarding not configured

Check IP forwarding:

```
sysctl net.ipv4.ip_forward # must return 1
```

If not 1, edit /etc/sysctl.conf, uncomment net.ipv4.ip_forward=1, then run:

```
sudo sysctl -p
```

Check routes:

```
ip route show
```

6.5.4 Error E4 — nftables rules missing or incorrect

The External Gateway requires all four of the following rules:

```
# inet filter table - chain forward:
iif "eth1" oif "eth0" accept
iif "eth0" oif "eth1" ct state established,related accept
iif "eth1" oif "eth1" accept

# ip nat table - chain postrouting:
oif "eth0" masquerade
```

Check with `sudo nft list ruleset`. If rules are missing, re-apply:

```
sudo nft -f /etc/nftables.conf
sudo systemctl restart nftables
```

Common mistake: masquerade on eth1 instead of eth0. The `oif` in the masquerade rule must always be eth0.

6.5.5 Error E5 — Cannot ping another VM

- Verify the target VM has the correct IP: `ip a`
- Check virtual switch assignments in Hyper-V (see Error E2)
- Confirm IP forwarding is enabled on both gateway VMs (see Error E3)
- Check source VM routes: `ip route show`
- Confirm `iif "eth1" oif "eth1" accept` exists in External Gateway forward chain (see Error E4)

6.5.6 Error E6 — No internet access

- Confirm masquerade rule is on eth0 — not eth1 (see Error E4)
- Confirm IP forwarding is enabled on External Gateway (see Error E3)
- Do not use ping to test internet — ICMP is blocked in this environment. The automarker uses HTTPS curl which is the correct test

6.5.7 Error E7 — nftables service not running

```
sudo systemctl enable --now nftables
sudo systemctl status nftables
```

6.6 Quick Reference

6.6.1 Error code summary

Code	Meaning	First command to run
E1	IP address wrong or missing	sudo nano /etc/netplan/50-cloud-init.yaml
E2	Network interface missing	Check Hyper-V Manager → VM Settings
E3	IP forwarding or route missing	sysctl net.ipv4.ip_forward
E4	nftables rules wrong or missing	sudo nft list ruleset
E5	Cannot ping another VM	Check E2, E3, E4 in order
E6	No internet access	Check E4 masquerade, then E3 forwarding
E7	nftables service not running	sudo systemctl enable --now nftables

6.6.2 Per-VM check summary

Check	Ext GW	Int GW	Server	Desktop
Interface existence	✓	✓	✓	✓
IP addressing	✓	✓	✓	✓
Default / static routes	✓	✓	✓	✓
IP forwarding	✓	✓	—	—
nftables service	✓	—	—	—
nftables forward rules	✓	—	—	—
nftables NAT masquerade	✓	—	—	—
Ping all reachable VMs	✓	✓	✓	✓
Internet access (HTTPS)	✓	✓	✓	✓